class 10th 20.3

Q4/

Find the value of m if,

i/ The roots of the eq/ $x^2-5x+2m=0$ differ by 1

Sol/

so According to given conditions.

The roots of the Given condition differ by 1.

which mean, ~~the roots~~ that the difference of roots equal to 1.

and roots are α and β so if roots are differ that mean $\boxed{\alpha-\beta=1 \rightarrow (i)}$

So

$x^2-5x+2m=0$

$a=1$

$b=-5$

$c=-2m$

so we know that $\alpha+\beta=-\frac{b}{a}$

$$=-\frac{(-5)}{1}=5$$

$$\boxed{\alpha+\beta=5}$$

and $\alpha\beta=\frac{c}{a}$

$$=\frac{2m}{1}=\boxed{\alpha\beta=2m}$$

so Now we have to find the $\alpha-\beta$, so In this case we have to use a such formula in which $(a-b)$ $(a+b)$ and (ab) should be included.

also we have formula:

$$(a+b)^2 - (a-b)^2 = 4ab$$

In this all the things are included there fore.

Now replace the value in αβ form:

$$(\alpha+\beta)^2 - (\alpha-\beta)^2 = 4\alpha\beta$$

Now just put the value which we have already been find out.

$$(\alpha+\beta)^2 - (\alpha-\beta)^2 = 4\alpha\beta$$

$$(5)^2 - (1)^2 = 4(2m)$$

$$25 - 1 = 8m$$

$$\frac{24}{8} = m$$

$$\boxed{3 = m}$$